Tutorial: exam-style practice

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

Part A: 10 multiple-choice questions (1 mark each). Part B: 4 structured questions (25 marks). Same format as the final exam.

Answer Part A by circling one option. Answer Part B in full, showing your working. The answer key is at the back.

Part A · Multiple choice

Choose one answer for each question.

A1 Topic: Introduction to Organic Molecule CH1-LO1

Which compound is **inorganic**? (Ball 12.E)

A. C_6H_{10}

B. $C_{12}H_{22}O_{11}$

C. $CoCl_2$

D. CH_3NH_2

A2 Topic: Introduction to Organic Molecule CH1-LO1

Which carbon compound is classed as **inorganic**?

A. CH_3OH

B. $CaCO_3$

C. C_2H_6

D. CH_3COOH

A3 Topic: Introduction to Organic Molecule CH1-LO1

A compound burns easily, melts at −20 °C and does not dissolve in water. It is most likely:

A. ionic
B. organic
C. a metal
D. a salt

A4 Topic: Introduction to Organic Molecule CH1-LO1

Carbon forms a huge number of compounds mainly because it:

A. forms four strong covalent bonds, including to other carbon atoms
B. is the most abundant element on Earth
C. forms ions easily
D. is a metal

A5 Topic: Introduction to Organic Molecule CH1-LO2

The molecular formula of $CH_3CH_2CH_2CH_3$ is:

A. C_4H_8
B. C_4H_{10}
C. C_3H_{10}
D. C_4H_{12}

A6 Topic: Introduction to Organic Molecule CH1-LO2

In a line-angle structure, which atoms are **not** written as symbols?

A. oxygen atoms
B. carbon atoms and the hydrogens on them
C. nitrogen atoms
D. chlorine atoms

A7 Topic: Introduction to Organic Molecule CH1-LO2

The H–C–H bond angle in methane is about:

A. 90°
B. 109.5°
C. 120°
D. 180°

A8 Topic: Introduction to Organic Molecule CH1-LO3

The functional group –COOH is called:

A. an ester
B. an aldehyde
C. a carboxylic acid
D. a ketone

A9 Topic: Introduction to Organic Molecule CH1-LO3

CH_3COCH_3 belongs to which family?

A. aldehyde
B. ketone
C. ether
D. ester

A10 Topic: Introduction to Organic Molecule CH1-LO4

Synthetic and natural samples of the same pure compound:

A. have different melting points
B. behave identically
C. have different formulas
D. differ in the number of bonds to carbon

Part B · Structured questions

Write your answers on paper, then self-mark with the worked solution.

Question B1 Topic: Introduction to Organic Molecule CH1-LO1 5 marks, C2

(a) Explain what *vitalism* was and how Wöhler's 1828 experiment challenged it. (3 marks)

(b) Give two reasons why carbon can form so many different compounds. (2 marks)

Question B2 Topic: Introduction to Organic Molecule CH1-LO1 6 marks, C4

Two white solids are labelled X and Y. X melts at 801 °C, does not burn and its aqueous solution conducts electricity. Y melts at 186 °C, chars and burns when heated, and its aqueous solution does not conduct.

(a) Classify X and Y as organic or inorganic, giving **two** reasons for each. (4 marks)

(b) State the type of bonding in each. (2 marks)

Question B3 Topic: Introduction to Organic Molecule CH1-LO2 7 marks, C3

For the compound $CH_3CH(CH_3)CH_2OH$:

(a) Write its molecular formula (1 mark)

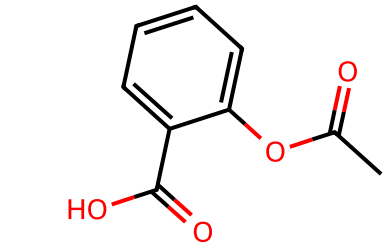
(b) Draw its expanded structural formula (2 marks)

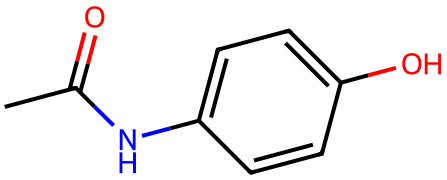
(c) Draw its line-angle structure (2 marks)

(d) State the shape and bond angle around the carbon that carries the CH_3 branch (2 marks)

Question B4 Topic: Introduction to Organic Molecule CH1-LO3 7 marks, C4

Aspirin and paracetamol are both used for pain and fever.





(a) Name all the functional groups in aspirin (top) and in paracetamol (bottom). (5 marks)

(b) A patient's aspirin tablets, kept in a humid bathroom, smell of vinegar. Explain why, naming the functional group involved. (2 marks)

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

Adapted from LibreTexts *Introduction to Organic Chemistry – Alkanes* 12.1, 12.2 and 12.4 (Sharpe Elles, from Ball, Hill & Scott) and *The Basics of GOB Chemistry* (Ball, Hill & Scott) 12.1, 12.4 and exercises 12.E, CC BY-NC-SA 3.0. Pharmacy items are original material by Dr. NAM. Shared under CC BY-NC-SA for non-commercial teaching.

PHD125 Pharmaceutical Organic Chemistry · Introduction to Organic Molecule

Tutorial: exam-style practice

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

Part A: 10 multiple-choice questions (1 mark each). Part B: 4 structured questions (25 marks). Same format as the final exam.

Answer key

Part A · Multiple choice

A1 Topic: Introduction to Organic Molecule CH1-LO1

Which compound is **inorganic**? (Ball 12.E)

A. C_6H_{10}

B. $C_{12}H_{22}O_{11}$

C. $CoCl_2$

D. CH_3NH_2

$CoCl_2$ contains no carbon.

A2 Topic: Introduction to Organic Molecule CH1-LO1

Which carbon compound is classed as **inorganic**?

A. CH_3OH

B. $CaCO_3$

C. C_2H_6

D. CH_3COOH

Carbonates are treated as inorganic by convention.

A3 Topic: Introduction to Organic Molecule CH1-LO1

A compound burns easily, melts at −20 °C and does not dissolve in water. It is most likely:

A. ionic

B. organic

C. a metal

D. a salt

Flammable, low-melting and insoluble in water: typical organic properties (Ball 12.E).

A4 Topic: Introduction to Organic Molecule CH1-LO1

Carbon forms a huge number of compounds mainly because it:

A. forms four strong covalent bonds, including to other carbon atoms

B. is the most abundant element on Earth

C. forms ions easily

D. is a metal

Catenation and tetravalency give carbon its diversity.

A5 Topic: Introduction to Organic Molecule CH1-LO2

The molecular formula of $CH_3CH_2CH_2CH_3$ is:

A. C_4H_8

B. C_4H_{10}

C. C_3H_{10}

D. C_4H_{12}

4 C and 3 + 2 + 2 + 3 = 10 H.

A6 Topic: Introduction to Organic Molecule CH1-LO2

In a line-angle structure, which atoms are **not** written as symbols?

A. oxygen atoms

B. carbon atoms and the hydrogens on them

C. nitrogen atoms

D. chlorine atoms

Ends and bends are carbons; H on carbon is implied.

A7 Topic: Introduction to Organic Molecule CH1-LO2

The H–C–H bond angle in methane is about:

A. 90°

B. 109.5°

C. 120°

D. 180°

Four electron groups → tetrahedral, 109.5°.

A8 Topic: Introduction to Organic Molecule CH1-LO3

The functional group –COOH is called:

A. an ester

B. an aldehyde

C. a carboxylic acid

D. a ketone

C=O with an OH on the carbonyl carbon.

A9 Topic: Introduction to Organic Molecule CH1-LO3

CH_3COCH_3 belongs to which family?

A. aldehyde

B. ketone

C. ether

D. ester

C=O between two carbons: a ketone (propanone, acetone).

A10 Topic: Introduction to Organic Molecule CH1-LO4

Synthetic and natural samples of the same pure compound:

A. have different melting points

B. behave identically

C. have different formulas

D. differ in the number of bonds to carbon

A compound is defined by its structure, not its source.

Part B · Structured questions

Question B1 Topic: Introduction to Organic Molecule CH1-LO1 5 marks, C2

(a) Explain what *vitalism* was and how Wöhler's 1828 experiment challenged it. (3 marks)

(b) Give two reasons why carbon can form so many different compounds. (2 marks)

(a) Vitalism was the belief that organic compounds could only be made by living organisms because they need a 'vital force'. Wöhler heated ammonium cyanate, an inorganic salt, and obtained urea, an organic compound found in urine. No living thing was involved, so organic compounds could be made in the laboratory.

(b) Any two: carbon forms four covalent bonds; it bonds strongly to other carbon atoms, forming chains, branches and rings; it bonds strongly to other elements such as H, O, N; it forms single, double and triple bonds.

Step	Marking criteria	Marks
1	Vitalism: organic compounds thought to need a vital force from living things	1
2	Wöhler made urea from ammonium cyanate (inorganic)	1
3	Conclusion: organic compound made without a living organism	1
4	First valid reason	1
5	Second valid reason	1
	Total	**5**

Question B2 Topic: Introduction to Organic Molecule CH1-LO1 6 marks, C4

Two white solids are labelled X and Y. X melts at 801 °C, does not burn and its aqueous solution conducts electricity. Y melts at 186 °C, chars and burns when heated, and its aqueous solution does not conduct.

(a) Classify X and Y as organic or inorganic, giving **two** reasons for each. (4 marks)

(b) State the type of bonding in each. (2 marks)

(a) X is inorganic: very high melting point; nonflammable; solution conducts (ions present). Y is organic: low melting point; flammable; solution does not conduct.

(b) X: ionic bonding (X is in fact NaCl). Y: covalent bonding (Y is sucrose).

Step	Marking criteria	Marks
1	X inorganic with first reason	1
2	X second reason	1
3	Y organic with first reason	1
4	Y second reason	1
5	X: ionic bonding	1
6	Y: covalent bonding	1
	Total	**6**

Question B3 Topic: Introduction to Organic Molecule CH1-LO2 7 marks, C3

For the compound $CH_3CH(CH_3)CH_2OH$:

(a) Write its molecular formula (1 mark)

(b) Draw its expanded structural formula (2 marks)

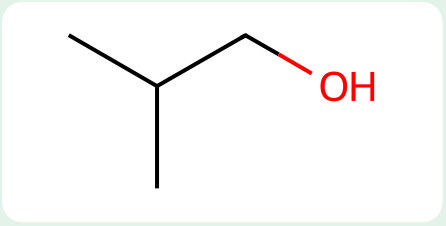
(c) Draw its line-angle structure (2 marks)

(d) State the shape and bond angle around the carbon that carries the CH_3 branch (2 marks)

(a) $C_4H_{10}O$.

(b) Four carbons: C1 (3 H) bonded to C2; C2 has 1 H and a CH_3 branch; C2 bonded to C3 (2 H), which carries the O–H. Every H drawn.

(c) A three-line 'Y' with one arm extended to end in OH:

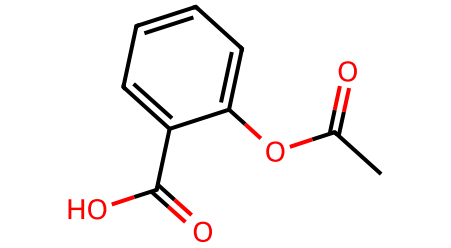


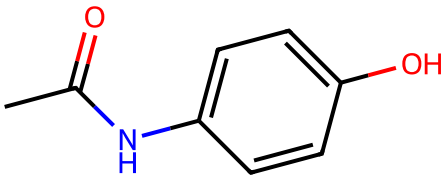
(d) That carbon has four single bonds: tetrahedral, about 109.5°.

Step	Marking criteria	Marks
1	$C_4H_{10}O$	1
2	Expanded: correct carbon skeleton with branch	1
3	Expanded: all 10 H and the O–H shown	1
4	Line: correct zig-zag with branch	1
5	Line: OH written, H on carbon omitted	1
6	Tetrahedral	1
7	109.5°	1
	Total	**7**

Question B4 Topic: Introduction to Organic Molecule CH1-LO3 7 marks, C4

Aspirin and paracetamol are both used for pain and fever.





(a) Name all the functional groups in aspirin (top) and in paracetamol (bottom). (5 marks)

(b) A patient's aspirin tablets, kept in a humid bathroom, smell of vinegar. Explain why, naming the functional group involved. (2 marks)

(a) Aspirin: aromatic ring, carboxylic acid, ester. Paracetamol: aromatic ring, phenol (–OH on the ring), amide.

(b) The ester group in aspirin is hydrolysed by moisture, forming salicylic acid and acetic acid; acetic acid gives the vinegar smell. The tablets should be discarded and medicines stored in a dry place.

Step	Marking criteria	Marks
1	Aspirin: aromatic ring	1
2	Aspirin: carboxylic acid and ester (both)	1
3	Paracetamol: aromatic ring	1
4	Paracetamol: phenol	1
5	Paracetamol: amide	1
6	Ester hydrolysed by water	1
7	Acetic acid formed (vinegar smell)	1
	Total	**7**

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

Adapted from LibreTexts *Introduction to Organic Chemistry – Alkanes* 12.1, 12.2 and 12.4 (Sharpe Elles, from Ball, Hill & Scott) and *The Basics of GOB Chemistry* (Ball, Hill & Scott) 12.1, 12.4 and exercises 12.E, CC BY-NC-SA 3.0. Pharmacy items are original material by Dr. NAM. Shared under CC BY-NC-SA for non-commercial teaching.